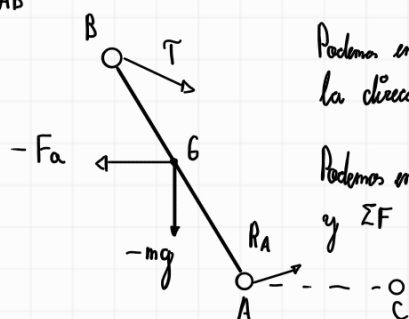
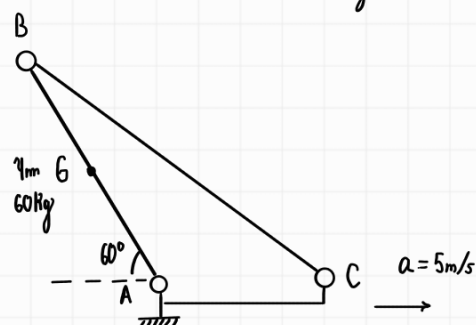
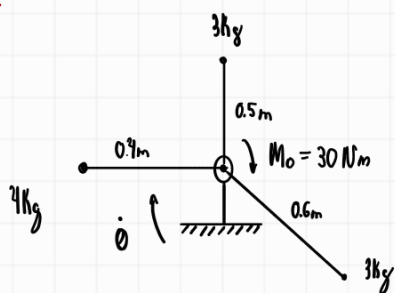


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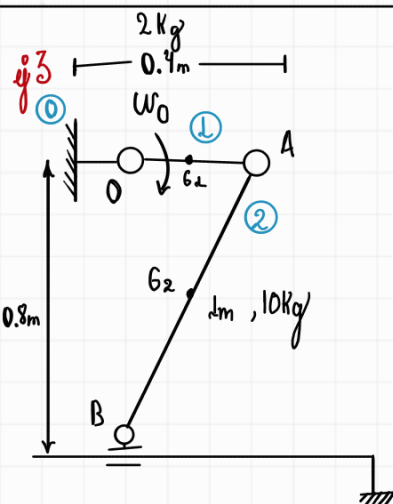
DINÁMICA DE SISTEMAS MECÁNICOS

PROBLEM SETS

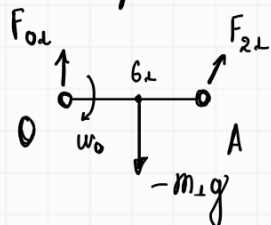


Podemos encontrar T con $\Sigma M_B = 0$ y la dirección de la cuerda

Podemos encontrar RA con EMA
y ΣF



DCL cuerpo 2

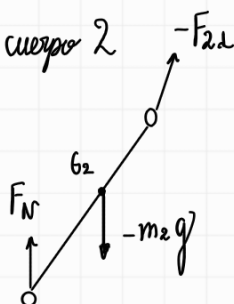


Para el cuerpo 1 las ecuaciones de N-E son

$$\Sigma \vec{F} = \vec{F}_{01} + \vec{F}_{21} + m_1 \vec{g} = m_1 \vec{a}_{G1}$$

$$\Sigma M_0 = W_0 + r_{0A} \times \bar{F}_{21} + r_{0G} \times (m_1 \bar{g}) = I_0 \alpha_L$$

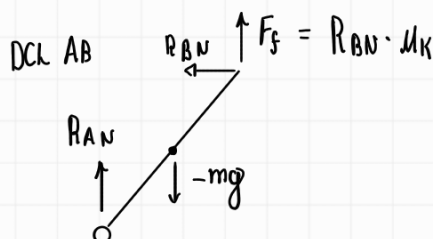
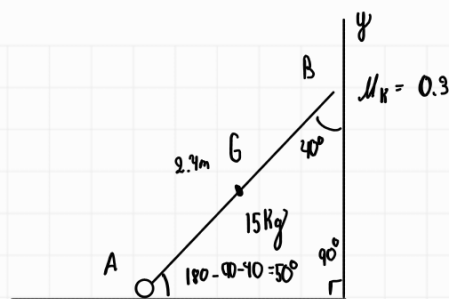
DCL cuerpo 2



Para el cuerpo 2 las ecuaciones de N-E son

$$\sum \vec{F} = \vec{F}_W - \vec{B} + m_2 \vec{g} = m_2 a_{B2}$$

$$\sum \vec{M}_B = \vec{r}_{BA} \times (-E_2) + \vec{r}_{BG_2} \times (m_2 \vec{g}) = I_{B_2} \alpha_2$$



Podemos hacer momentos en B y después RA, luego solo quedarán dos incógnitas

$$\Sigma F_x = R_{BN} = -m a_{Gx}$$

$$\sum F_y = F_f + R_{AN} - mg = ma_{cy}$$

$$\Sigma M_A = r_{AG} \times -mg + r_{AB} \times -R_{BN} + r_{AB} \times F_f = I_A \alpha$$

$$I_A = \frac{1}{3} m L^2$$

$$Q_{Ax} = Q_{Gx} + \alpha \times r_{GA}$$

$$Q_{By} = Q_{By} + \alpha \times r_{GB}$$